Product Rule for Bilinear Forms

- $b : \mathcal{U} \times \mathcal{V} \to \mathbb{R}$ is **Bilinear** if for all $\alpha, \beta \in \mathbb{R}$, $\boldsymbol{u}, \boldsymbol{u}_1, \boldsymbol{u}_2 \in \mathcal{U}$, and $\boldsymbol{v}, \boldsymbol{v}_1, \boldsymbol{v}_2 \in \mathcal{V}$

$$b(\alpha \boldsymbol{u}_1 + \beta \boldsymbol{u}_2, \boldsymbol{v}) = \alpha b(\boldsymbol{u}_1, \boldsymbol{v}) + \beta b(\boldsymbol{u}_2, \boldsymbol{v})$$
$$b(\boldsymbol{u}, \alpha \boldsymbol{v}_1 + \beta \boldsymbol{v}_2) = \alpha b(\boldsymbol{u}, \boldsymbol{v}_1) + \beta b(\boldsymbol{u}, \boldsymbol{v}_2)$$

- For differentiable $\boldsymbol{u}(t)$ and $\boldsymbol{v}(t)$ in finite dimensional vector spaces $\mathcal{U}, \mathcal{V}$

$$\boxed{\frac{d}{dt} b(\boldsymbol{u}(t), \boldsymbol{v}(t)) = b(\boldsymbol{u}'(t), \boldsymbol{v}(t)) + b(\boldsymbol{u}(t), \boldsymbol{v}'(t))}$$

- Write Δ for the change from t to $t + h$. Bilinearity gives

$$\frac{\Delta b}{h} = b\left(\frac{\Delta \boldsymbol{u}}{h}, \boldsymbol{v}(t+h)\right) + b\left(\boldsymbol{u}(t), \frac{\Delta \boldsymbol{v}}{h}\right)$$

Take $h \to 0$. Finite dimensional linear forms are continuous.